

# AGENTIC RETRIEVAL AUGMENTED GENERATION: A MODERN APPROACH TO KNOWLEDGE SYSTEMS

## Abstract

Agentic Retrieval-Augmented Generation (Agentic RAG) combines retrieval systems with reasoning agents to improve factual accuracy and reduce hallucinations.

## Introduction

Traditional language models often generate responses without verifying information sources. Agentic RAG addresses this challenge by retrieving relevant documents before generating answers.

## Methodology

The system consists of:

- Vector Database
- Retrieval Engine
- Large Language Model
- Citation Generation Module

## Results

Agentic RAG improves answer accuracy, transparency, and explainability compared to standard language model workflows.

## Conclusion

Future research may focus on multi-agent retrieval systems and adaptive reasoning strategies.